

Symbol	Parameter	Conditions		Min	Typ	Max	Unit
f _{SCK}	SPI clock frequency	Slave receiver mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	100	MHz
		Slave transmitter mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	-	30	
t _{su(NSS)}	NSS setup time	Slave mode		3	-	-	ns
t _{h(NSS)}	NSS hold time	Slave mode		1	-	-	
t _{w(SCKH)}	SCK high and low time	Master mode		T _{SCK} ⁽¹⁾ - 1	T _{SCK} ⁽¹⁾	T _{SCK} ⁽¹⁾ + 1	
t _{w(SCKL)}							
t _{su(MI)}	Data input setup time	Master mode	DRDS = 0	3	-	-	
			DRDS = 1	13 - T _{SCK} /2 ⁽¹⁾			
Slave mode		2	-	-			
t _{su(SI)}							
t _{h(MI)}	Data input hold time	Master mode	DRDS = 0	1.5	-	-	
			DRDS = 1	T _{SCK} /2 ⁽¹⁾ - 8	-	-	
Slave mode		1.5	-	-			
t _{h(SI)}							
t _{a(SO)}	Data output access time	Slave mode		11	13.5	16.5	
t _{dis(SO)}	Data output disable time	Slave mode		9.5	11.5	14	
t _{v(SO)}	Data output valid time	Slave mode, 2.7 V ≤ V _{DD} ≤ 3.6 V		-	12	16.5	
t _{v(MO)}	Data output valid time	Master mode		-	1.5	2	
t _{h(SO)}	Data output hold time	Slave mode (after enable edge)		8	-	-	
t _{h(MO)}	Data output hold time	Master mode (after enable edge)		0	-	-	

1. $T_{SCK} = T_{PCLK} \times \text{Baud rate prescaler}/2$.

Figure 37. SPI timing diagram - slave mode and CPHA = 0

